| Name: | Class: | Date given: |
|---|---|---|
| | | Date due in: |

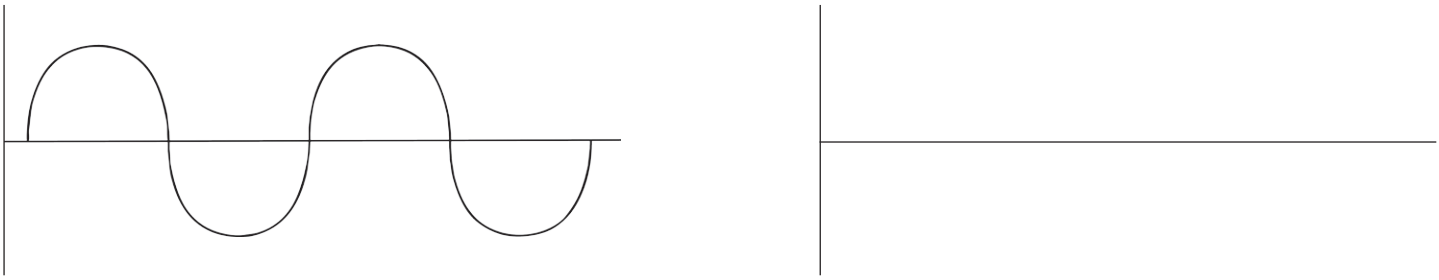
# Sound Waves

1. Choose the correct word from the box below to match to each of the descriptions. [3]

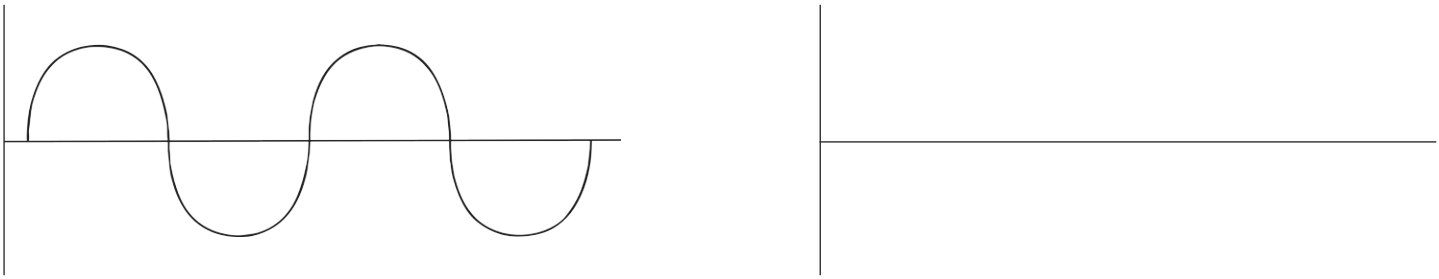
| |
|---|
| pitch |
| frequency |
| amplitude |

| |
|---|
| The number of waves passing a given point each second. |
| The tone of a sound (how high or low it is). |
| The loudness or volume of a sound. |

2. Look at the first sound wave. Draw another sound wave which would have a lower pitch. [1]



3. Look at the first sound wave. Draw another sound wave which would have a louder sound. [1]



4. Explain why sounds waves cannot travel through space. [3]



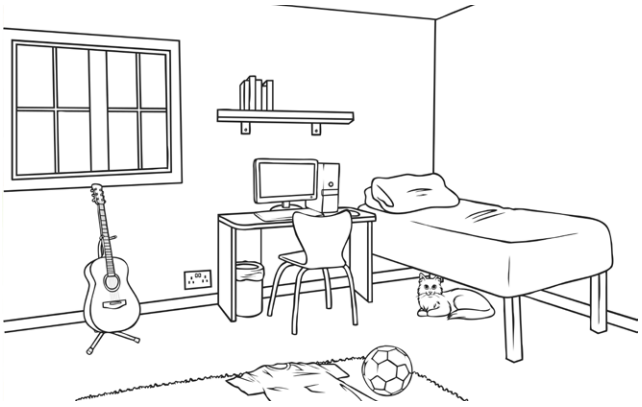
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5a. Using your understanding of how sound waves travel through different mediums, write the most appropriate wave speed for each medium into the table. [3]

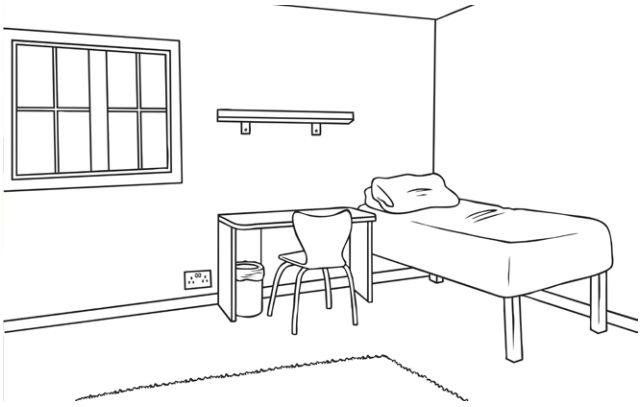
343 m/s 1493 m/s 5130 m/s

| Medium | Wave speed |
|---|---|
| Steel | |
| Water | |
| Air | |

5b. Some students measured the speed of sound in air by clapping and measuring the time taken to hear the echo. What other piece of information would they need in order to calculate wave speed? [1]



Room A



Room B

6a. Look at the pictures of the two rooms above. Which room would produce the least echoes? [1]

6b. Explain your answer. [2]

**Learning Outcomes** (tick if achieved)

| | | |
|---|---|---|
| Q1, 2, 3 | I can describe properties of sound waves | |
| Q4, 5 | I understand how sound travels through a medium | |
| Q6 | I can describe reflection and absorption of sound waves | |



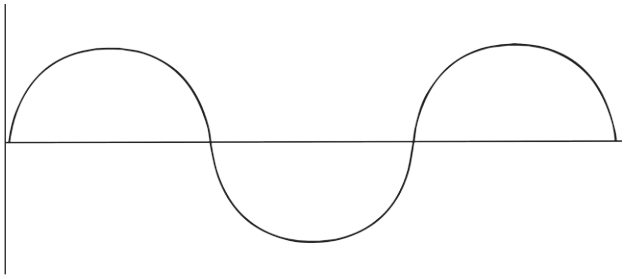
visit twinkl.com

# Sound Waves **Answers**

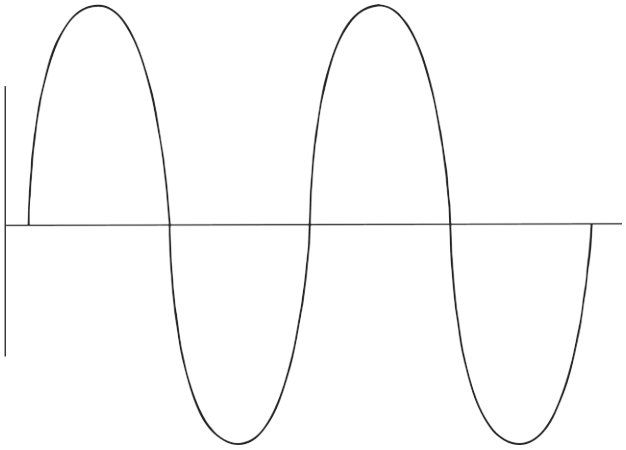
1. Choose the correct word from the box below to match to each of the descriptions. [3]

| | |
|---|---|
| pitch | The number of waves passing a given point each second. |
| frequency | The tone of a sound (how high or low it is). |
| amplitude | The loudness or volume of a sound. |

2. Look at the first sound wave. Draw another sound wave which would have a lower pitch. [1]



3. Look at the first sound wave. Draw another sound wave which would have a louder sound. [1]



4. Explain why sounds waves cannot travel through space. [3]

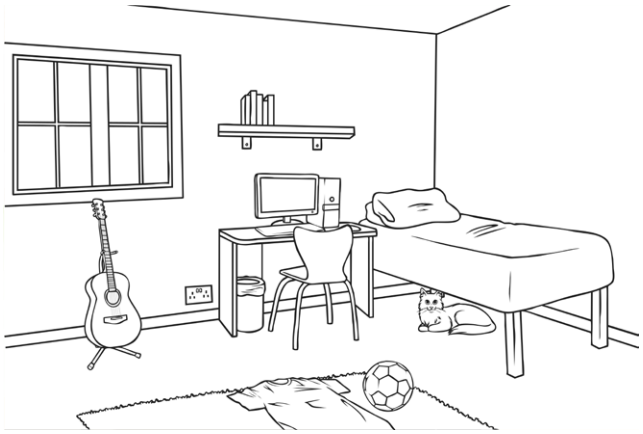
- **sound travels as vibrations**
- **particles are needed to transfer the vibrations**
- **space is a vacuum (no particles)**

5a. Using your understanding of how sound waves travel through different mediums, write the most appropriate wave speed for each medium into the table. [3]

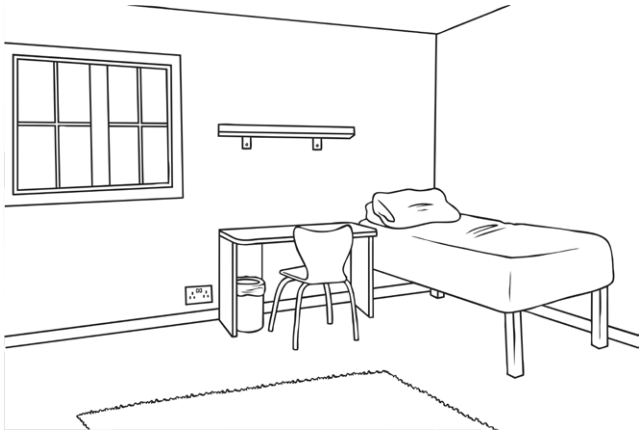
| Medium | Wave speed |
|---|---|
| Steel | **5130 m/s** |
| Water | **1493 m/s** |
| Air | **343 m/s** |

5b. Some students measured the speed of sound in air by clapping and measuring the time taken to hear the echo. What other piece of information would they need in order to calculate wave speed? [1]

**distance travelled (metres)**



Room A



Room B

6a. Look at the pictures of the two rooms above. Which room would produce the least echoes? [1]

**Room A**

6b. Explain your answer. [2]

- **an echo is caused by sound being reflected from smooth surfaces**
- **carpets / curtains etc in room A will absorb the sound rather than reflecting it**